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Dynamic V2X Perception from Road-to-Vehicle Vision

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Abstract*—Vehicle-to-everything (V2X) perception is an innovative technology that enhances vehicle perception accuracy, thereby elevating the security and reliability of autonomous systems. However, existing V2X perception methods exhibit limitations at traffic intersections due to a failure to account for effective collaboration between dynamic vehicles and stable Roadside Unit (RSU) in adapting to traffic environments. To harness the use of RSU and adapt V2X perception models to dynamic scenes, we propose an *Adaptive Road-to-Vehicle Perception* (AR2VP) approach, building V2X perception based on road-to-vehicle visual information. In AR2VP, we leverage RSU to offer stable, wide-range sensing capabilities and serve as communication hubs. AR2VP is devised to tackle both intra-scene and inter-scene changes. For the intra-scene changes, we construct a Dynamic Perception Representation (DPR) module, which effectively integrates vehicle perceptions, enabling vehicles to capture a more comprehensive range of dynamic factors within the scene. Moreover, we introduce a Road-to-Vehicle Perception Compensating (R2VPC) module, aiming at preserving the maximized RSU perception information in the presence of intra-scene changes. For the inter-scene changes, we implement a Prompt-Replay mechanism leveraging the RSU's limited storage capacity to retain a subset of typical historical scene samples, maintaining model robustness in response to inter-scene changes. We conduct comprehensive experiments on 3D object detection and segmentation, and the results show that AR2VP excels in both performance-bandwidth trade-offs and adaptability within dynamic environments. Our code is available at https://github.com/tjy1423317192/AR2VP.

***Index Terms*—V2X, Roadside unit, Perception, Dynamic scene, Continual Learning**

I. Introduction

Environmental perception [2] is important for autonomous vehicles [19, 28, 35], enabling them to express, identify, and interpret sensory inputs to understand complex environments.

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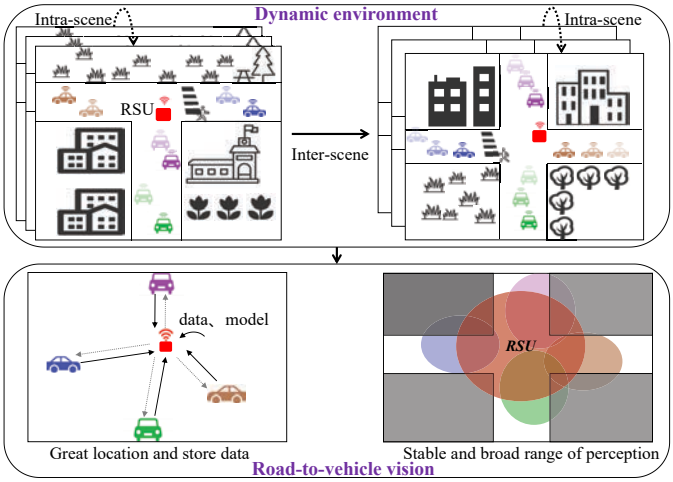

Fig. 1: Dynamic Scene Advantage of RSU. RSU demonstrates stability in perception and possesses geographical advantages in intra-scene changes, as opposed to the continual mobility of vehicles in inter-scene dynamics. With storage and communication capabilities, RSU stores old scenes data and model, enabling adaptation to inter-scene changes.

In recent years, there have been in-depth researches on single-vehicle perception [29], with researchers proposing various algorithms for handling different downstream tasks. Despite progress in single-vehicle perception, individual perspectives often lead to a decline in perception capability in distant or obstructed areas. Vehicle-to-everything (V2X) technology [8, 17, 27, 46] is an advanced communication system facilitating dialogue between vehicles and other entities [15, 52], including vehicle-to-vehicle and vehicle-to-infrastructure communication, single-vehicle perception can be upgraded to cooperative perception. V2X communication introduces additional perspectives that assist autonomous vehicles in seeing farther, clearer, and even subtler obstacles, thereby enhancing perception capabilities.

While current V2X methods [20, 21] showcase commendable perception capabilities in autonomous driving on straight roads, its performance falters in complex traffic intersections—precisely the locations where traffic accidents are prone to occur. The challenging task in real-world environmental perception of traffic intersections comprises two key aspects: (1) *Intra-scene variations*: This involves the variations within a scene, containing factors like pedestrians in motion and vehicles executing turns. (2) *Inter-scene variations*: This pertains

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to substantial changes in architecture and lane configurations between different scenes. We argue that the reason for the failure of the current V2X methods in complex intersections is rooted in the over-reliance on the vehicle perception vision. Only aggregating vehicles in unpredictable locations may overlook some important scene elements around intersections, such as the pedestrians crossing streets. In general, these methods lack the effective use of static and stable visual information from the RSU. On top of this, the RSU [18] serves as an important support for the V2X system, because a stationary RSU positioned at a fixed location, provides a more stable and expansive sensory coverage while minimizing redundant communication. RSU facilitates the collection of comprehensive perception data over a wider range and ensures more consistent perception as compared to inherently mobile vehicles. Furthermore, RSU can exploit the increased storage and computing capabilities, to adapt to the wide range of inter-scene variations.

Nevertheless, training V2X using RSU to achieve effective perception in complex dynamic intersections is not a trivial issue. The primary challenge lies in how to adapt to the aforementioned two dynamic scenarios. On one hand, the locations of smart vehicles at each intersection are always uncertain, making it difficult to effectively integrate vehicle perception. The dynamicity of vehicles often complicates it even further to evaluate the performance of a V2X system [34]. As the positions and directions of vehicles constantly change, it is challenging to predict how they will affect traffic flow and safety. To achieve comprehensive perceptions of the dynamic traffic environment, it is necessary to collect data from vehicles at different angles and positions and analyze their data together to form a complete understanding of the entire intersection. On the other hand, existing have not fully considered the practical cross-scene learning process, resulting in perception models frequently experiencing abrupt errors in this process and lacking adaptability. In different traffic scenarios, such as busy city center intersections and suburban intersections with less traffic, perception models need to be able to adapt to these different environments and conditions. In this paper, we focus on optimizing multi-perspective perception fusion within scenes and addressing cross-scene adaptability issues at intersections with the use of RSU, as existing technologies frequently struggle with transitioning between these scenarios, resulting in perception errors or failures.

Motivated by this, this paper proposes an **Adaptive Road-to-Vehicle Perception (AR2VP)** approach, which constructs a road-to-vehicle cooperative perception model tailored for dynamic environments. As shown in Fig. 1, within dynamic traffic scenarios, RSU serves as the perception nodes situated at the heart of the traffic scene, equipped with additional storage space and computational capabilities. The geographical advantage of these units allows them to process and analyze traffic data more effectively. In each specific traffic scenario, RSU shows outstanding sensing capabilities, enabling them to capture and analyze the complexity of traffic flow from multiple perspectives. Our work re-explores the latent advantages of the RSU and leverages the road-to-vehicle vision paradigm to enhance V2X perception within dynamic scenes. First, to effectively handle intra-scene changes, we design DPR module and R2VPC module. These modules collaboratively construct an adaptable graph catering to intra-scene changes, thereby enhancing vehicles' overall adaptability within dynamic scenarios. Second, to effectively adapt to inter-scene changes, we present RSU-Prompt Replay (RPR), utilizing RSU limited storage capacity and typical sample replay techniques in continual learning [24, 25], enabling vehicles to adapt to large-scale scene transitions beyond intra-scene changes. Our approach is validated on the tasks of scene segmentation and 3D object detection. The results on the V2X-Sim dataset [17] show good perception performance and the adaptability of AR2VP to dynamic scenes.

Our contributions are three-fold:

(1) We propose an AR2VP approach for dynamic V2X perception to address V2X perception challenges at complex traffic intersections in dynamic scenarios. AR2VP effectively coordinates perceptual information between vehicles and RSU at traffic intersections and applies to optimizing perception across different traffic intersections.
(2) We design DPR module and R2VPC module to deal with intra-scene changes. DPR leverages the geographical advantage and perceptual insights from roadside locations to establish an effective fusion of vehicle-road information. Meanwhile, R2VPC enhances the overall adaptability of vehicles in dynamic scenarios by improving the stability of roadside perception.
(3) We present RPR strategy to effectively handle inter-scene changes. Prompt-replay utilizes the perceptual completeness of updated RSU features to train learnable scene prompts, summarizing information about the entire intersection scene. This allows for the extraction of a small number of representative samples that characterize the intersection, enabling high model robustness with a limited RSU storage capacity.

II. Related Work

A. Perception in V2X

V2X technology encompasses various forms of cooperative communication, including Vehicle-to-Vehicle (V2V) [6, 7] and Vehicle-to-Infrastructure (V2I) [41]. All agents engage in mutual communication on the broadcast platform. The core step in V2X perception technology involves the adoption of a cooperative strategy. Current approaches mainly include raw measurement-based early collaboration, output-based late collaboration, and feature-based intermediate fusion.

Early fusion [3] integrates raw measurements from all agents, providing a comprehensive perspective to effectively address occlusion and long-range issues in single-agent perception. However, it requires substantial communication bandwidth. *Late fusion* combines each agent's perception outputs, emphasizing bandwidth efficiency. Yet, this approach may result in noisy and incomplete individual perception outputs, leading to suboptimal fusion results. To strike a balance between performance and bandwidth, *intermediate Fusion* [20, 21] is introduced. This strategy aggregates intermediate

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features across agents [26], potentially achieving both communication bandwidth efficiency and enhanced perception ability. Nevertheless, insufficient collaboration strategy design may lead to information loss during feature abstraction and fusion, limiting improvements in perception ability.

For intermediate fusion technology, Who2com [21] utilizes a handshake communication mechanism to determine which two agents should be used for image segmentation, providing a simple yet effective communication approach. However, this method may be limited by considering communication only between two agents and may not be flexible enough in larger-scale scenarios. When2com [20] introduces an asymmetric attention mechanism to decide when to communicate and how to create communication groups for image segmentation. This method makes communication between agents more flexible, allowing for automatic adjustment of communication frequency as needed. However, due to the introduction of more parameters and complexity, it may require more computational resources and longer training times in practical applications. V2VNet [42] proposes multi-round message passing on a spatial-aware graph neural network for joint perception and prediction in autonomous driving. This method fully leverages the spatial relationships between vehicles and enables end-to-end learning in perception and prediction tasks, thereby improving overall performance. However, due to the need for larger computational overhead and more data to train complex neural network models, it may require more time and resources. DiscoNet [15] introduces a distilled collaboration graph with matrix-valued edge weights for adaptive perception, offering a superior performance-bandwidth trade-off. This method reduces communication overhead by using matrix-valued edge weights, thus reducing bandwidth requirements while maintaining high performance. However, the need for graph simplification and optimization may increase some computational complexity and training costs. Vadivelu, Ren, Tu, Wang, and Urtasun [37] proposes a pose error regression module to learn to correct pose errors when the pose information from other vehicles is noisy. For *V2I* [11] technology, the collaboration is between infrastructure and vehicles, which expands the vehicle's perception field. Overall, these methods have their strengths and weaknesses. Who2com is straightforward but may be limited by small-scale communication. When2com algorithm offers greater flexibility but may require more computational resources. V2VNet maximizes spatial information but may require more data and time for training. DiscoNet provides a better balance between performance and bandwidth but may require more computational complexity. Compared to earlier research, these intermediate collaboration algorithms propose new methods for handling communication and cooperation in perception tasks, thus advancing the field in collaborative approaches and information fusion.

However, most of V2V and V2I methods build V2X perception model from vehicle vision, which is insufficient in the dynamic traffic environment.[33, 34, 44] In this paper, we aim to construct a road-to-vehicle vision to address the challenge of inadequate adaptability of collaborative perception models in dynamic environments.

B. Continual Learning

Continual learning [51] is a commonly employed approach to adapt to constantly changing scenarios. It allows the model to continuously accept new data while updating itself, even when old data samples are scarce or non-existent, preventing the forgetting of memories associated with the old data. This learning method is more intelligent and practical, especially suitable for situations where, due to limitations such as hardware or privacy reasons, it is not possible to retain a large amount of old data in its entirety. (See Fig. 2).

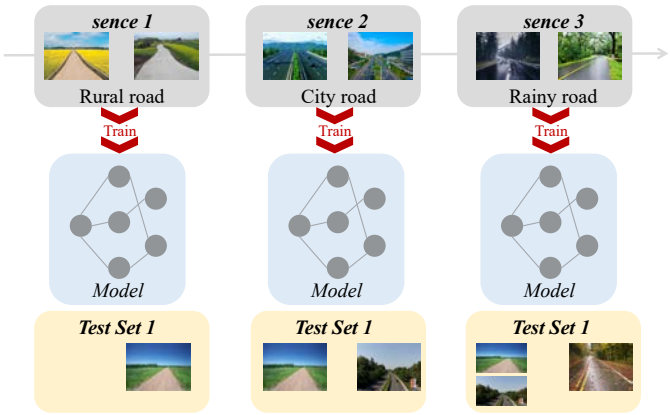

Fig. 2: The setting of continual learning. Non-overlapping scene data arrives sequentially, and the model needs to learn how to perceive all scenes. After the learning process for each task, the model is evaluated across all learned scenes. An ideal model should perform well on newly learned scenes and remember previous scenes without catastrophic forgetting.

Therefore, in cross-scenario learning, traffic scenes exhibit notable variability. To address conflicts between old and new knowledge, the model needs to continually learn from diverse and ever-changing new data, enabling vehicles to better adapt to various environmental changes. Continuous learning holds promise for application in complex and dynamic traffic environments. In such scenarios, some common continual learning methods include regularization [9, 12]: A technique to prevent overfitting by adding penalties for complex model structures, experience replay [32, 38, 40]: Periodically training with past data to retain knowledge and balance learning, and parameter freezing [36, 49]: Keeping certain model parameters fixed to preserve existing knowledge while adapting to new data.

This paper found through experiments that traditional V2X technologies overlook the stability issue in cross-scene learning (no retesting of old scene data with high scene differences in the experimental settings), leading to forgetting phenomena [45**?**] in perception models when regressing old scene data. Therefore, This paper investigates the potential of applying continual learning to model V2X perception systems, with the aim of better adapting to inter-scene changes, thereby enhancing the safety and reliability of the automomous.

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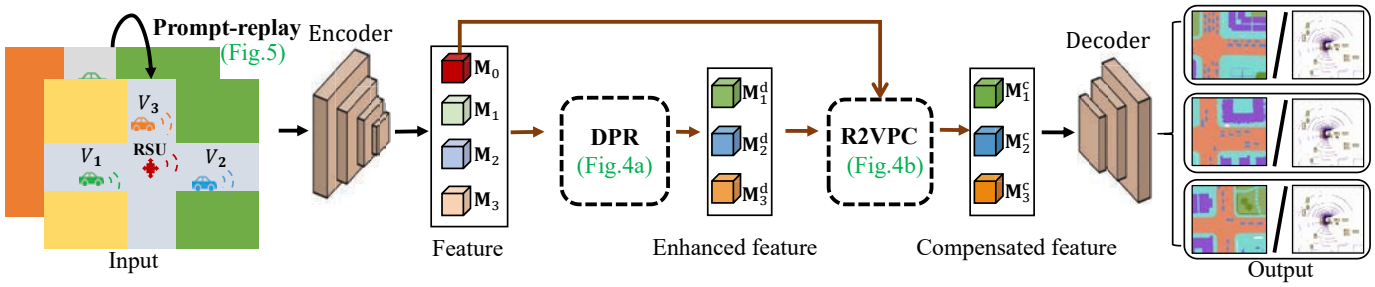

Fig. 3: Overall perception model framework of AR2VP. For the vehicle V1 (green) as an example, its BEV map $\mathbf{V}_1$ is encoded by the shared Encoder to obtain the feature map $\mathbf{M}_1$. Based on the DPR module, the neural information from other vehicles and RSU is aggregated to obtain the feature map $\mathbf{M}_1^{\mathrm{d}}$. It is compensated with the neural information $\mathbf{M}_0$ from the RSU to obtain the feature map $\mathbf{M}_1^{\mathrm{c}}$. The shared header outputs the result after the shared Decoder. Best view in color.

III. Adaptive Road-to-Vehicle Perception

A. Overview

We study the V2X perception task with RSU placed on dynamic intersection scenes. In any scene, the V2X perception consists of an RSU and vehicles. The RSU and vehicles collect point cloud data, these input single-view point cloud can be converted to bird's-eye-view (BEV) [38] maps $\mathcal{V} = \{\mathbf{V}_0, \mathbf{V}_1, \mathbf{V}_2, ..., \mathbf{V}_i\}$, where $\mathbf{V}_0$ for RSU and $\mathbf{V}_i$ for the i-th vehicle ($i > 0$). Existing V2X perception technologies struggle to effectively coordinate complex dynamic traffic data [34], which falls short of meeting the safety requirements in dynamic traffic environments: Intra-scene changes, such as pedestrians in motion and moving vehicles, along with inter-scene changes like transitions between extensive structures and road layouts across different locations, introduce disruptions to V2X perception, potentially compromising vehicle safety.

Motivated by this, this work considers building a collaborative perception model from road-to-vehicle vision for sensing complex and dynamic traffic scenarios. We name the method Adaptive Road-to-Vehicle Perception (AR2VP), as shown in Fig. 3, where vehicles and RSU communicate and cooperate through a broadcast communication channel. AR2VP considers to address two kinds of scene changes:

(1) **Intra-scene changes:** we first design *DPR* module, utilizing RSU geographical and perceptual advantages to effectively integrate the perception from vehicles, enabling vehicles to capture a more comprehensive range of dynamic factors within the scene. Then, to further enhance vehicles perception capabilities in dynamic environments, we draw inspiration from residual [23, 31] techniques and propose *R2VPC* module. Leveraging RSU perceptual advantages, this module compensates the post-collaborative perception of vehicles, filling in intra-scene dynamic factors that are overlooked by the vehicles, thereby further enhancing the overall adaptability of vehicles to dynamic environments. Lastly, it enables vehicles to adapt to large-scale scene transitions beyond intra-scene changes.

(2) **Inter-scene changes:** we introduce RSU-Prompt Replay, leveraging the perceptual completeness of RSU to train learnable scene prompts for summarizing information about the entire intersection scene. This allows RSU to extract a small number of representative samples that characterize the intersection, enabling effective suppression of scene distribution drift with limited RSU storage capacity. This adaptation ensures that AR2VP can robustly perceive vehicles while accommodating variations between scenes.

B. Overcoming Intra-Scene Changes

1) Dynamic Perception Representation: Vehicle perception varies with the changes of dynamic entities within the intra-scene. To effectively coordinate these dynamic factors, this paper proposes DPR module, which constructs a directed collaborative graph $\mathcal{G} = \{\mathcal{M}, \xi\}$ that leverages the advantages of RSU to adapt to intra-scene changes (See Fig. 4(a)), where $\mathcal{M} = \Phi_{\text{shared}}(\mathcal{V})$ is encoded by the shared encoder $\Phi_{\text{shared}}(\cdot)$ to generate the feature maps and $\mathcal{V}$ represents BEV maps. The collaborative graph has three stages for dynamic perception representation:

(1) Stage S1: position information transforming. In this stage, each vehicle transfers the position to the RSU for interaction. RSU and each vehicle has its own independent position $\mathcal{P} = \{(x_0, y_0), (x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$. When selecting the position of the i-th vehicle for collaboration, we need to transform the position information of RSU (x_0, y_0) into $(x_{0\to i}, y_{0\to i})$ corresponding to the i-th vehicles using the position matrix:

$$x_{0\to i} = \mathbf{R}_i \mathbf{R}_0^\top (x_0 - x_i), \quad y_{0\to i} = \mathbf{R}_i \mathbf{R}_0^\top (y_0 - y_i) \tag{1}$$

where $\mathbf{R}_i$ denotes the rotation matrix of the i-th vehicle, which represents the orientation of the coordinate system relative to the reference coordinate system. Note that $\mathbf{R}_0$ is RSU's rotation matrix. Then, we pass the converted position information to the S2 stage to obtain the edge weights.

(2) Stage S2: position-guided feature fusing. In this stage, each vehicle receives effective perception information from both RSU and other vehicles in the same scene. RSU, due to its unique geographical position, offers vehicles dynamic environmental adaptability. Therefore, based on the position information from Stage S1, we combine the relative distance between RSU and vehicles with the feature information between vehicles and carry out effective collaborative perception. In the directed collaborative graph $\mathcal{G}$, to determine edge

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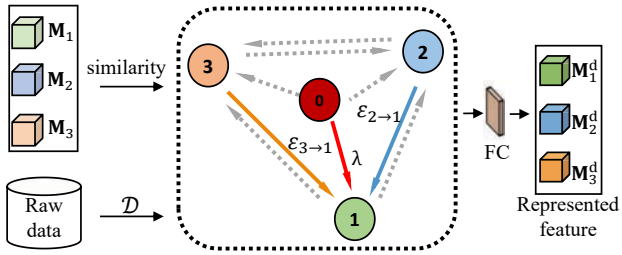
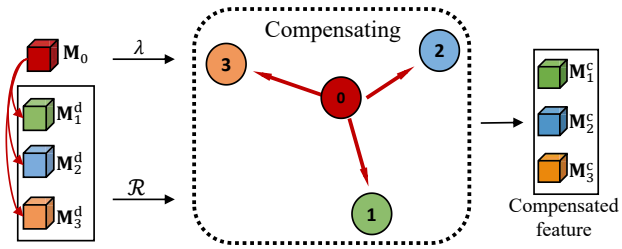


(a) Dynamic Perception Representation. For the feature map $\mathbf{M}_1$, we calculate the weights ξ by combining feature similarity with the RSU-Vehicle distances $\mathcal{D} = \{d_1, d_2, ..., d_i\}$ to construct a directed collaborative graph $\{\mathcal{M}, \xi\}$, followed by perception representing of the feature map $\mathbf{M}_i^{\mathrm{d}}$.



(b) Road-to-Vehicle Perception Compensating. For the feature map $\mathbf{M}_1^{\mathrm{d}}$, utilizing RSU-vehicles similarity ratio $\mathcal{R}$ and RSU perception threshold λ to determine the extent of RSU perception compensation within $\mathbf{M}_1^{\mathrm{d}}$, thus obtaining the compensated feature map $\mathbf{M}_1^{\mathrm{c}}$.

Fig. 4: Dynamic Perception Representation and Road-to-Vehicle Perception Compensating.

weights ξ, we firstly obtain the distances $\mathcal{D} = \{d_1, d_2, ..., d_i\}$ between vehicles and the RSU from the Stage S1:

$$d_i = \sqrt{(x_{0\to i} - x_i)^2 + (y_{0\to i} - y_i)^2}. \tag{2}$$

Then, we associate the features of different vehicles. In other words, the matrix value of the edge weight from the 2nd vehicle to the 1st vehicle $\xi_{2\to 1}$:

$$\xi_{2\to 1} = \frac{d_2 \cdot \cos(\mathbf{M}_1, \mathbf{M}_2)}{\sum_{i=2}^{N} d_i \cdot \cos(\mathbf{M}_1, \mathbf{M}_i)} \tag{3}$$

where $\mathrm{norm}(\cdot)$ represents set normalization, and $\cos(\cdot)$ represents feature similarity.

For the edge weights between RSU and vehicles, we use a fixed weight $\lambda = \frac{1}{N}$ to retain the stable perception information of RSU, and N represents the number of agents. The theoretical foundation for pre-setting λ is inspired by multi-agent coordination techniques [50], where key agents act as coordinators. In this paper, RSU plays a role highly similar to that of key agents, serving as coordinators in the information collaboration process. Thus, we use $\frac{1}{N}$ as a coordination parameter, aiming to provide certain fairness in collaboration weight. Through Stage 2, we complete the construction of graph $\mathcal{G}$.

(3) Stage S3: feature information aggregating: In this stage, we utilize the directed collaborative graph constructed in Stage S2 to synergistically enhance the representation of each vehicle. The perception information of each vehicle and RSU is integrated to better capture dynamic entities, achieving a comprehensive perception of the entire environment. Specifically, each vehicle aggregates the normalized edge-weighted features of all other vehicles. The updated feature map of the i-th vehicle is $\hat{\mathbf{M}}_i$:

$$\hat{\mathbf{M}}_i = \sum\nolimits_{j=1}^{N} \xi_{j\to i}\mathbf{M}_i + \lambda\mathbf{M}_0. \tag{4}$$

In the DPR module, this study leverages the perceptual and geographical advantages of RSU to assist vehicles in perception fusion. This approach enables the perception model to initially adapt to dynamic environments, achieving a comprehensive perception effect.

2) Road-to-Vehicle Perception Compensating: Due to the continuous changes of scenes, using only the collaborative graph for vehicle perception in dynamic scenarios is insufficient. This paper further leverages the advantages of RSU perceptual stability and extensive coverage to compensate for the updated vehicles perception, thereby enhancing vehicles perception in dynamic scenes. At this stage, our objective is to utilize the perception features of RSU to compensate for the updated feature maps of vehicles during the decoding process (See Fig. 4(b)).

First, we flat the feature maps of the RSU and vehicles:

$$\mathbf{F}^i = \mathrm{flatten}(\mathbf{M}_i^{\mathrm{d}}) \tag{5}$$

where $\mathbf{M}_i^{\mathrm{d}}$ is obtained by decoded from $\hat{\mathbf{M}}_i$ using a fully-connected layer. Then, To compute the unique linear similarity between different vehicles and RSU $\mathcal{R} = \{r_1, r_2, ..., r_i\}$, considering the partial overlap in the perspectives of different vehicles, we employed the Pearson correlation coefficient calculation method [30]). Before the calculation, we centralized the data to eliminate the translational effect of redundant data, thereby emphasizing the unique correlation between features more prominently, which is to accurately determine which feature map needs to be compensated using RSU perception, avoiding unnecessary computational burdens. The formula is as follows:

$$r_i = \frac{\sum_{i=1}^{n} a_i}{\sqrt{\sum_{i=1}^{n} a_0^2 \cdot a_i^2}} \tag{6}$$

where

$$a_i = \sum\nolimits_{j=1}^{n} (\mathbf{F}_j^i - \sum\nolimits_{i=1}^{k} \frac{\mathbf{M}_i^d}{k}). \tag{7}$$

Then, a predefined threshold λ is used to determine whether compensation with RSU is required for feature mappings. We supplement them with RSU perception information, resulting in compensated feature maps $\mathcal{M}^{\mathrm{c}}$:

$$\mathbf{M}_i^{\mathrm{c}} = (\lambda - r_i)\mathbf{M}_0^{\mathrm{d}} + \mathbf{M}_i^{\mathrm{d}}. \tag{8}$$

Finally, the model is followed by different output heads based on the tasks (such as segmentation (seg.) and detection (det.)) to generate perception results. Though AR2VP adapts to the changes of intra-scene, adapting to the changes of inter-scene is still unsolved.

C. Overcoming Inter-Scene Changes

Due to the relatively abrupt changes of inter-scene, models often face the challenge of distributional adaptation drift when learning new environments [22, 53]. In this situation, the model suffers from serious catastrophic forgetting [10, 47],

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Algorithm 1 Intra-scene adaptation

Input: BEV maps $\mathcal{V}$, position $\mathcal{P}$
Output: Compensated feature maps $\mathcal{M}^{\mathrm{c}}$

1: Encoder($\mathcal{V}$)$\rightarrow\mathbf{M}$ *#Agent features*
2: $\mathcal{P}\rightarrow$ Eq. (2) $\rightarrow\mathcal{D}$ *#Vehicle-RSU distances*
3: $(\mathbf{M},\mathcal{D})\rightarrow$ Eq.(3) $\rightarrow\xi_{i\rightarrow j}$*#Correlations between agents*
4: $(\xi_{i\rightarrow j},\mathbf{M})\rightarrow$ Eq.(4) $\rightarrow\hat{\mathcal{M}}$ *#Update features*
5: $\hat{\mathcal{M}}\rightarrow$ Eq.(5) and (6) $\rightarrow\mathcal{R}$ *#Feature similarity ratio*
6: $(\mathcal{R},\hat{\mathcal{M}})\rightarrow$ Eq.(8) $\rightarrow\mathcal{M}^{\mathrm{c}}$ *#Improve vehicle perception with RSU*
7: **return** $\mathcal{M}^{\mathrm{c}}$

which may lead to the risk of traffic accidents. Inspired by the experience replay in continual learning, we propose the RPR method for V2X perception. RPR takes advantage of the unique perceptual completeness and stability of RSU in the scene. RPR effectively suppresses distribution adaptation drift and reduces catastrophic forgetting, but it also optimizes the storage requirements that replay methods entail.

In the RPR module, we seek to utilize limited storage of RSU to reduce the drift as much as possible. Firstly, we build prompts to extract representative frames in the V2X perception, by encapsulating the perceptual information of the entire global scene. Specifically, we first randomly initialize k prompts $\mathcal{P}=\{\mathbf{P^1},\cdots,\mathbf{P^k}\}$. Then, we update prompts using the Exponential Moving Average (EMA) algorithm [1] to learn various aspects of continuous scene data and reduce the potential impact of errors from a single prompt.

$$\mathbf{P_t^i}=\alpha\mathbf{S_t}+(1-\alpha)\mathbf{P_{t-1}^i}\quad t=1,\cdots,n \tag{9}$$

where, $\alpha=0.5$ represents the smoothing factor. To make the RSU feature computed with the $\mathcal{P}$, we pass $\mathbf{M}_0$ through the fully connected layer f_n to get $\mathbf{S_t}$,

$$\mathbf{S_t}=f_n(\mathbf{M}_0), \tag{10}$$

it signifies the features coalesced by RSU in the t-th frame of the scene. Employing the EMA algorithm helps decrease the accumulation of redundant information within prompts, providing a more comprehensive representation of overall scene information.

Subsequently, while learning in new scenes, based on these prompts, typical scene samples $\mathcal{S}_{\mathrm{p}}$ are sliced and stored in RSU:

$$\mathcal{S}_{\mathrm{p}}=\mathcal{S}_{\mathrm{c}}[\mathrm{top}_l(\cos_t(\mathbf{P_t},\mathbf{S}_t))]\quad t=1,2,3,... \tag{11}$$

where, $\mathcal{S}_{\mathrm{c}}$ represents new scene data. $\mathrm{top}_l(\cdot)$ represents extracting the indices of the top l samples with higher global information. Therefore, different typical samples can be selected based on the similarity of the prompts.

Finally, when the model learns a new scenario again, $\mathcal{S}_{\mathrm{p}}$ and model will be updated, and effective knowledge review will be achieved by revisiting these stored representative samples. This method can not only effectively alleviate scene distribution drift and reduce model forgetting, but also optimize RSU storage capacity to achieve minimum storage and robust forgetting suppression:

$$\theta=\mathrm{SGD}(\mathcal{S}_{\mathrm{c}}\cup\mathcal{S}_{\mathrm{p}},\theta). \tag{12}$$

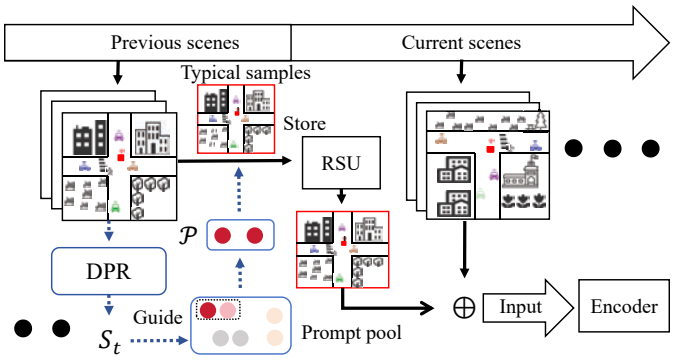

Fig. 5: Prompt-replay. After calculation by the DPR model, the feature information $\mathbf{S_t}$ of RSU of each frame is obtained, and then $\mathbf{S_t}$ is used to guide the training of prompt $\mathcal{P}$ to participate in the extraction of typical samples in the previous scenes. Finally, the Current scenes and typical samples are input into the encoder for model training.

Algorithm 2 Inter-scene adaption

Input: RSU feature $\mathbf{S}_t$, new scene data $\mathcal{S}_{\mathrm{c}}$
Output: A few typical scene data $\mathcal{S}_{\mathrm{p}}$

1: **if** The first scene **then**
2: $(\mathbf{S}_t,\mathcal{S}_{\mathrm{c}})\rightarrow$ Eq. (9) $\rightarrow\mathbf{P}$ *#Update prompts*
3: $(\mathbf{P},\mathbf{S}_t)\rightarrow$ Eq. (11) $\rightarrow\mathbf{sim}$ *#Calculate similarity*
4: $\mathcal{S}_{\mathrm{c}}\rightarrow$ Eq. (11) $\rightarrow\mathcal{S}_{\mathrm{p}}$ *#Extract typical samples*
5: **else**
6: $(\mathcal{S}_{\mathrm{c}},\mathcal{S}_{\mathrm{p}})\rightarrow$ Eq. (13) $\rightarrow\mathcal{S}_{\mathrm{p}}$ *#Update typical samples*
7: **end if**
8: **return** $\mathcal{S}_{\mathrm{p}}$

This formula represents model update, where $\mathrm{SGD}(\cdot)$ is the stochastic gradient descent algorithm, and θ represents the model parameters. And the next formula represents $\mathcal{S}_{\mathrm{p}}$ update:

$$\mathcal{S}_{\mathrm{p}}=S_{\mathrm{c}}[\mathrm{top}_l(\cdot)]\cup\mathcal{S}_{\mathrm{p}}. \tag{13}$$

In this module, the model engages in comprehensive learning of the previous typical data stored in RSU and the current scene data. This not only acquires knowledge from new scenes but also revisits knowledge from previous scenes, achieving a synergistic combination of learning and review. Our Prompt-replay effectively mitigates the catastrophic forgetting caused by inter-scene changes.

D. The Whole Algorithm

In the process of model learning update to dataset $\mathcal{S}$, we use L_{det} loss for the detection [14, 39] task to update learning:

$$L_{\mathrm{det}}=\sum\nolimits_{i=1}^{n}\frac{\eta(Y_i-Y_i')^2}{\sigma^2} \tag{14}$$

where η typically takes a value of 0.5 and 0.7.

In the segmentation [5, 48] task, we use L_{seg} loss for update learning:

$$L_{\mathrm{seg}}=-\sum\nolimits_{i=1}^{n}(Y_i\cdot\log(Y_i')) \tag{15}$$

where Y and Y' represent the label and prediction in scene $\mathcal{S}$, σ is a hyperparameter.

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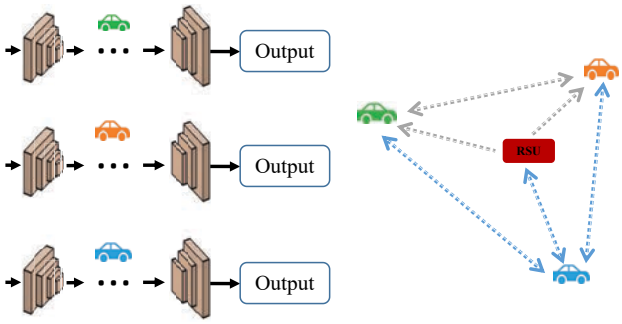
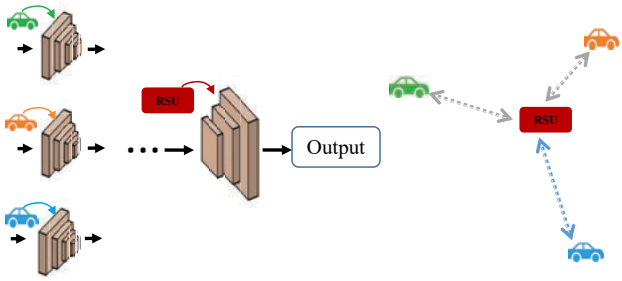
Algorithm 3 AR2VP

Input: New scene data $\mathcal{S}_{\mathrm{c}} \subset (\mathcal{V}, \mathcal{P})$
Output: Prediction Y_i'

for each scene $i \in [0, n]$ **do**
  if $i = 0$ **then**
    $\mathcal{S}_{\mathrm{c}} \rightarrow$ *Alg. 1* $\rightarrow \mathcal{M}^{\mathrm{c}}$ *#Get compensated feature maps*
    Alg. 2 $\rightarrow \mathcal{S}_{\mathrm{p}} \Rightarrow RSU$ *#Store typical samples*
  else
    $RSU \Rightarrow \mathcal{S}_{\mathrm{p}}$ *#Get typical samples*
    $(\mathcal{S}_{\mathrm{p}}, \mathcal{S}_{\mathrm{c}}) \rightarrow$ *Alg. 1* $\rightarrow \mathcal{M}^{\mathrm{c}}$
    Alg. 2 $\rightarrow \mathcal{S}_{\mathrm{p}} \Rightarrow RSU$ *#Update and store typical samples*
  end if
  $\mathcal{M}^{\mathrm{c}} \rightarrow Y_i'$ *#Get prediction*
end for
return Y_i'



(a) When RSU lacks additional computational power. Vehicles will store the entire model. One vehicle communicates $2n$ times, and the entire collaborative system requires communication in the order of $2n^2$ times.



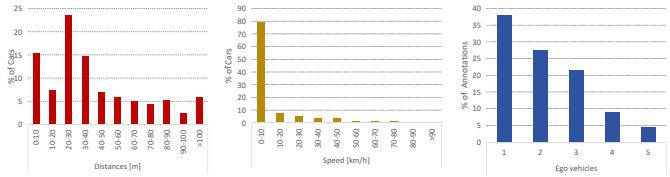
(b) When RSU has additional computational power. Vehicles will only store the encoder model. This will significantly reduce communication between vehicles. One vehicle communicates 2 times, and the entire collaborative system requires communication in the order of $2n$ times.

Fig. 6: The storage method of the model is closely related to the number of communications. If RSU has additional computational power, the number of communications will be reduced by $2n^2 - 2n$ times (in an ideal scenario).

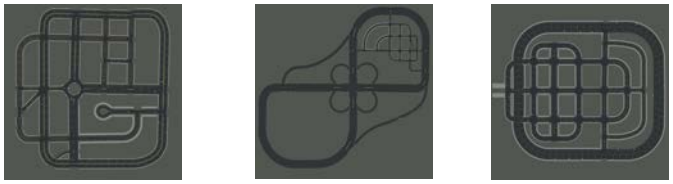
The overall updated loss L of the model is as follows:

$$L = L_{\mathrm{det/seg}}^{\mathrm{previous}} + L_{\mathrm{det/seg}}^{\mathrm{current}} \tag{16}$$

where $L_{\mathrm{det/seg}}^{\mathrm{previous}}$ represents the loss from the replayed data in previous scenes for the current task (det./seg.), and $L_{\mathrm{det/seg}}^{\mathrm{previous}}$ represents the loss from current scene data used for the current task.



(a) Left: Statistics of the distance between every two ego vehicles for all frames. Middle: Speed of cars located within 70m from ego vehicles. Right: Percentage of annotated vehicles observed by 1-5 ego vehicles.



(b) The map of towns which are squared-grid towns with cross junctions and multiple lanes per direction [13].

Fig. 7: Dataset illustration.

In the whole AR2VP research (See Fig. 3 and Algorithm 3), we design the DPR module (See Fig. 4(a) and Algorithm 1), merging geographical and feature data from RSU and vehicles to create an adaptable collaborative graph for dynamic scenarios. This effectively integrates perception information from different vehicles, enabling a more comprehensive grasp of dynamic elements within the scene. Subsequently, inspired by residual techniques, we propose the R2VPC module (See Fig. 4(b) and Algorithm 1). By leveraging RSU perceptual advantages, this module compensates post-collaborative vehicle perception, filling in intra-scene dynamic elements overlooked by the vehicles, further enhancing overall adaptability to dynamic settings. Lastly, to extend adaptability beyond intra-scene changes, we introduce Prompt-replay (See Fig. 5 and Algorithm 2), in conjunction with RSU perceptual completeness and continual learning techniques, to alleviate the issue of distribution adaptation drift within the model. This empowers AR2VP to cope with inter-scene changes, ensuring robust and reliable vehicle perception. In general, in the context of intra-scene dynamic perception learning, we employ the full-scenario training method, exclusively utilizing Algorithm 1. When dealing with inter-scene transformation perception learning, we adopt the cross-scene learning training approach, incorporating both Algorithm 1 and Algorithm 2.

E. Bandwidth Discussion

We speculate and analyze the vehicle-road cooperative communication in two scenarios based on the presence or absence of additional computational power in RSU, as shown in Fig. 6.

(1) If RSU has normal computational capabilities (containing only encoders), vehicles communicate with each other. Communication details are outlined in Fig. 6(a).

(2) If RSU has extra computational power (including decoders), we store the entire model in the RSU. This exchange of additional RSU computational power is used to gain advantages in vehicle-road cooperative communication, as detailed in Fig. 6(b). Therefore, regarding the two scenarios mentioned

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TABLE I: Detection performance in comparison with other methods.

Method	AP(%) @IoU=0.5	AP(%) @IoU=0.7
Early Fusion	96.63	96.05
Late Fusion	85.62	83.84
When2com	81.35	80.02
When2com*	81.86	80.69
Who2com	81.32	79.98
Who2com*	81.69	80.66
V2V	91.89	89.90
Disco	92.01	90.41
AR2VP	**94.50**	**92.77**

above, we can draw conclusions. Storing the model in the RSU further reduces n^2 communication bandwidth.

AR2VP can be optimized not only by leveraging the additional computational power of RSU to increase or reduce communication bandwidth but also by optimizing data transmission to save the communication bandwidth. Since AR2VP employs an intermediate fusion approach, the data transmitted in terms of communication consists of feature information, where RSU and vehicles could compress their feature map prior to transmission. Optionally, in our study, we make use a 1 × 1 convolutional autoencoder [4] to compress and decompress the feature maps along the channel dimension. The autoencoder is trained together with the whole system. Therefore, AR2VP can increase communication efficiency by a factor of n through feature compression, where n represents the degree of feature compression. We show this in our experiments (see the subsection IV-F).

IV. Experiment

A. Data Preparation

In this study, In this study, we evaluated V2X perception tasks offline using the V2X-sim dataset. The V2X-sim dataset emulates multi-agent scenarios, wherein each scenario encompasses a 20-second traffic flow across multiple intersections. Laser radar recordings are captured at intervals of 0.2 seconds, yielding a total of 100 frames per scenario. This dataset comprises 100 distinct scenes, with each frame housing multiple samples. The training set comprises 23,500 samples, while the test set contains 3,100 samples. To establish a fixed large scenario, we selected 30 scenes, which collectively contribute 3,000 frames. Among these, the training set comprises 2,700 frames, while the test set consists of 300 frames. Moreover, to implement cross-scene experiments, we also train V2X model sequentially on three major scenes in chronological order. Fig. 7(a) shows the details of the dataset. and Fig. 7(b) shows three major scenes.

B. Evaluation Metric

In this paper, we evaluate our method on two V2X perception tasks, including scene segmentation and vehicle objection. We employ the generic BEV detection evaluation metric: *Average Precision* (AP) [15] at *Intersection-over-Union* (IoU) threshold of 0.5 and 0.7:

$$\mathrm{AP} = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN})) \tag{17}$$

where True Positives (TP) refers to the number of correctly detected objects by the model. This is determined by comparing the predicted bounding boxes with the ground truth bounding boxes in terms of their overlap. If the predicted bounding box has sufficient overlap with any of the ground truth bounding boxes, it is considered a true positive. False Positives (FP) refers to the number of objects incorrectly detected by the model, such as identifying background or non-existing objects as targets. This is determined by examining the overlap between predicted bounding boxes and all ground truth bounding boxes. If the overlap between the predicted bounding box and all ground truth bounding boxes is insufficient, it is considered a false positive. False Negatives (FN) refers to the number of real objects missed by the model. If the ground truth bounding box of an object has no overlap with any predicted bounding box or has insufficient overlap, it is considered a false negative.

We evaluate the segmentation performance using mean IoU (mIoU). We evaluate the extent of forgetting across inter-scene changes using Forget. *IoU* threshold of 0.5 and 0.7:

$$\mathrm{mIoU} = \sum\nolimits_{\mathrm{i}=1}^{\mathrm{N}} (\mathrm{IoU_i})/\mathrm{N} \tag{18}$$

where $\mathrm{IoU_i}$ is the IoU for each agent. N is the total number of agents.

C. Compared Methods

We first compare with the early fusion method [3] and the Late fusion method. Furthermore, four intermediate fusion

TABLE II: Segmentation comparison of intra-scene. (The best options in intermediate collaboration methods are shown in bold and $*$ indicates the pose-aware version.)

Method	Unlabeled	Vehicles	Sidewalk	Ground	Road	Buildings	Pedestrian	Vegetation	mIoU(%)
Early Fusion	65.96	90.87	94.67	94.52	97.37	94.89	50.45	90.26	84.87
Late Fusion	48.56	72.17	86.88	85.96	93.48	85.92	18.21	80.07	71.41
When2com	41.25	65.47	69.62	58.83	83.65	62.36	27.18	62.00	58.79
When2com*	41.42	63.47	72.19	58.81	81.02	68.55	28.18	74.36	59.75
Who2com	42.25	66.47	70.62	59.83	84.65	63.36	28.18	63.00	59.80
Who2com*	40.02	63.47	72.60	62.81	81.00	60.55	28.20	66.36	60.75
V2V	60.10	84.92	93.04	91.87	95.98	**93.10**	33.89	86.85	79.01
Disco	61.15	84.75	92.82	92.62	96.52	92.95	**35.49**	**87.01**	80.41
AR2VP	**98.89**	**85.31**	**93.37**	**92.86**	**96.63**	93.31	33.63	86.38	**85.05**

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methods are used, inculuding When2com [20] which uses asymmetric attention for image segmentation communication. Who2com [21], V2V [43] which employs spatial-aware graph neural networks for autonomous driving perception. and Disco [16] which offers adaptive perception with a matrix-weighted collaboration graph. Since the original Who2com and When2com do not consider pose information, we consider both pose-aware and pose-agnostic versions (with $*$) to achieve fair comparisons. All the methods use the same segmentation and detection backbones and conduct collaboration at the same intermediate feature layer.

We utilize data stored in RSU (MB) to mitigate the forgetting issue caused by inter-scene variations. These stored data consist of two types: firstly, the original old scene data from RSU-Random Replay (RRR) methods, and secondly, the prompt data proposed in RSU-Prompt Replay (RPR). Leveraging these lightweight Prompt data can to some extent substitute the old scene data, thereby reducing storage capacity and computational burden.

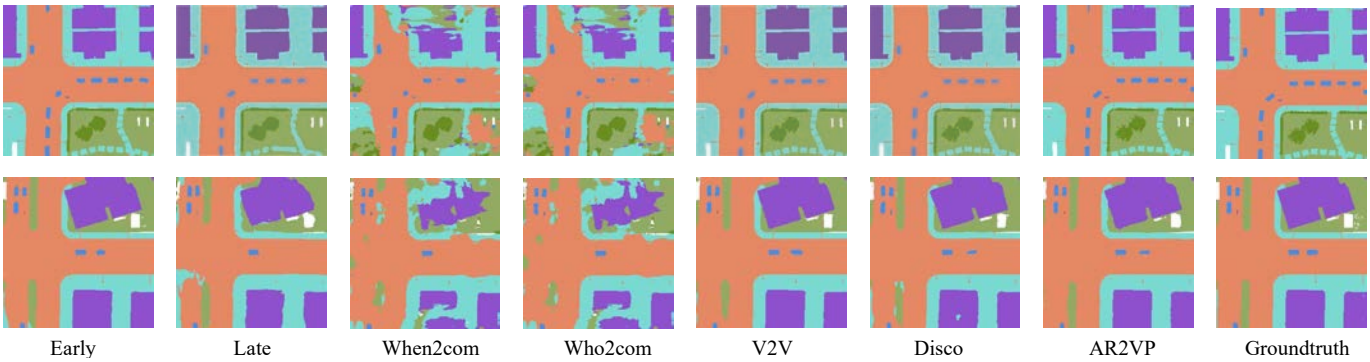
D. Comparisons under Intra-scene Changes

Tables II and I show the comparisons in terms of mIoU (seg.) and AP@0.5/0.7 (det.). Compared to the pose-aware When2com, AR2VP improves by 57.57% in segmentation of unlabeled data and 26.26% in mIoU. Compared to Disco, AR2VP improves by 4.64%. Compared to the pose-aware When2com, AR2VP improves by 13.15% in AP@0.5 and 12.75% in AP@0.7. Compared to Disco, AR2VP improves by 3.4% in AP@0.5 and 3.2% in AP@0.7. The qualitative results are shown in Fig. 8(a) (seg.) and Fig. 8(b) (det.).

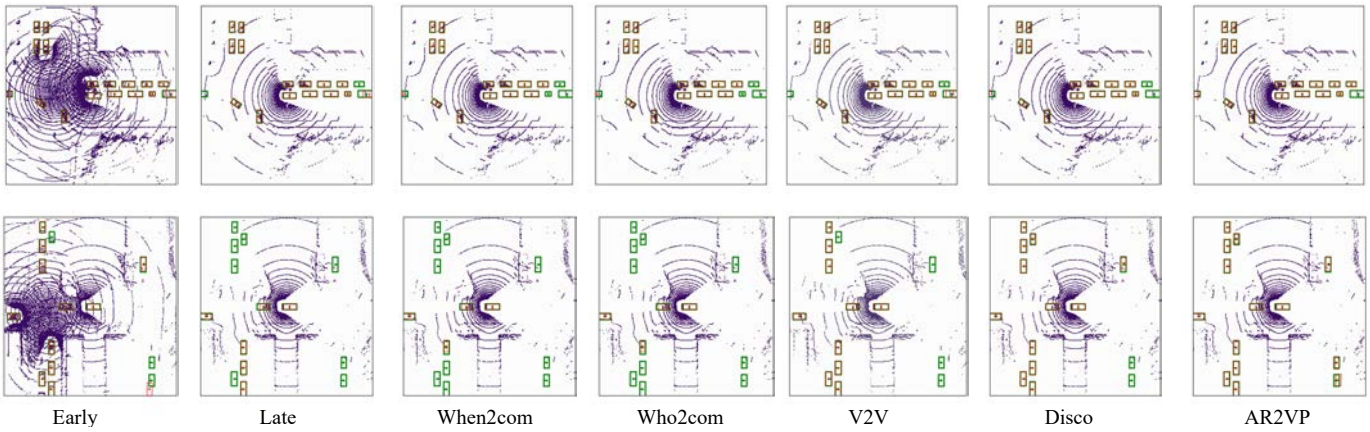
We observed that AR2VP demonstrates superior entity perception outcomes, achieving the highest overall perception performance. This analysis underscores that current V2X technologies rarely rely on RSU to expand perception horizons. In contrast, AR2VP harnesses the latent strengths of RSU to address intra-scene changes, which enhances the vehicle's ability to adapt to dynamic scenes, consequently elevating the overall perception capabilities. However, AR2VP does exhibit a performance drawback in pedestrian detection, implying a particular challenge in detecting small targets.

E. Comparisons under Inter-Scene Change

Table III shows the comparison on inhibition of forgetting across different scenes. Compared to the pose-aware When2com, AR2VP improves by 30.86% in mIoU and reduces the forgetting rate by 20.32%. Compared to Disco, AR2VP improves by 22.08% in mIoU and reduces the forgetting rate by 24.34%; Compared to the pose-aware When2com, AR2VP improves by 30.27% in AP@0.5 and 28.38% in AP@0.7, and reduce forgetting rate by 26.07% in AP@0.5 and 25.60% in AP@0.7. Compared to Disco, AR2VP improves by 6.05% in AP@0.5 and 6.44% in AP@0.7 and reduces the



(a) Visualizations of collaborative BEV semantic segmentation.



(b) Visualizations of BEV detection on V2X-Sim. Red and green boxes are the predictions and ground-truths respectively.

Fig. 8: Visualizations of collaborative BEV.

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TABLE III: Perception comparisons on inter-scene changes. (RSU-Random Replay (RRR) represents the random extraction of old scene samples, RSU-Prompt Replay (RPR) represents the extraction of typical samples using prompts.)

Method	Det.(@IoU=0.5)		Det.(@IoU=0.7)		Seg.		Mem.(MB)↓
	AP(%)↑	Forget(%)↓	AP(%)↑	Forget(%)↓	mIoU(%)↑	Forget(%)↓	
Early Fusion	85.15	14.96	81.03	19.19	52.71	42.62	0
Late Fusion	62.77	27.61	57.86	31.96	4014	31.91	0
When2com	61.40	32.23	57.90	34.44	40.73	30.76	0
When2com*	61.64	31.38	57.98	33.01	45.60	28.65	0
Who2com	61.04	32.89	57.66	34.63	41.63	31.56	0
Who2com*	61.47	32.01	58.11	33.79	44.71	29.56	0
V2V	82.54	15.65	76.98	20.63	48.50	46.08	0
Disco	83.47	14.39	78.46	18.78	48.09	43.31	0
V2V(RRR)	87.63	7.38	83.48	9.46	63.06	18.09	240
Disco(RRR)	87.65	7.61	83.65	9.53	64.74	19.12	240
AR2VP(RRR)	89.52	6.16	84.90	8.84	71.51	11.48	240
AR2VP(RPR)	**90.98**	**5.11**	**86.56**	**7.92**	**74.13**	**10.10**	240
AR2VP(RPR)	90.23	5.71	85.78	8.45	73.66	10.54	**130**

TABLE IV: Ablation studies of intra-scene.

RSU	**graph**	**compensator**	Det.(%)		Seg.(%)
			Ap@0.5	Ap@0.7	mIoU
✗	✗	✓	66.99	65.47	55.01
✗	✓	✗	89.88	87.95	75.65
✗	✓	✓	90.65	88.46	73.06
✓	✗	✓	68.56	66.32	56.47
✓	✓	✗	93.80	91.71	84.04
✓	✓	✓	**94.50**	**92.77**	**85.05**

TABLE V: Ablation studies of inter-scene.

Prompt	Seg.(%)		
	80(MB)	130(MB)	240(MB)
2	67.98	70.86	72.42
3	**68.71**	**73.66**	74.13
4	68.58	73.53	**74.62**

forgetting rate by 8.23% in AP@0.5 and 9.94% in AP@0.7. Furthermore, AR2VP, when utilizing Prompt-replay, maintains robust perceptual capabilities even when subjected to minimal usage of 110MB, outperforming RSU-Random Replay (RRR) in similar conditions.

AR2VP presents itself as a frontrunner in terms of overall perception performance. Upon analysis, it's evident that traditional V2X technologies disregard the influence of inter-scene changes on perception. In contrast, AR2VP exploits the storage capacity of RSU and integrates continual learning principles to effectively address inter-scene changes. This strategic approach empowers vehicles to assimilate new scenes while minimizing the extent of memory loss from prior scenes. This capability shows a strong adaptability to inter-scene changes in perception, thereby enhancing the global robustness of perception.

The prompt integrates RSU features across different time steps to represent dynamic panoramic information, thereby enhancing the existing perception system's resistance to forgetting. Compared to other methods, APVP focuses more on utilizing RSU perception data for dynamic scene understanding. Therefore, although Prompt-replay can be applied to other models to mitigate forgetting, AR2VP notably demonstrates the most favorable suitability. Fig. 9 shows the learning processes of new scene data based on the old model, revealing the degree of memory forgetting from previous scenes. The observation is clear: V2V and Disco struggle to accommodate inter-scene changes, leading to significant memory loss from previous scenes. In contrast, AR2VP adeptly navigates inter-scene changes, exhibiting a higher retention of memories from prior scenes. This analysis underscores AR2VP's capacity for the lowest forgetting rate and the most proficient performance in addressing inter-scene changes.

F. Performance-Bandwidth Trade-off Analysis

In Fig. 10, we compare the proposed AR2VP with the baseline methods in terms of the trade-off between segmentation performance and communication bandwidth. The dashed line represents the baseline based on when2com. To show the better trade-off of the proposed AR2VP, we employ an autoencoder to compress features and reduce the communication bandwidth used for feature transmission (AR2VP$/x$ means compress x times). Compared to AR2VP, AR2VP/32 degrades by 1.31% in segmentation, still outperforming Disco in terms of mIoU. This means the compressed features in AR2VP do not significantly compromise perception performance.

G. Ablation Study

We conduct ablation studies to analyze the perceptual performance of the graph and the communication in the presence and absence of RSU. The results are shown in Table IV. Firstly, we find that the participation of RSU in the collaborative process provides additional perception coverage to enhance vehicle perception performance. Secondly, the collaborative graph effectively integrates all perception information, enabling vehicles to comprehensively perceive entities within the scene. Finally, in scenarios where RSU is present, the compensator utilizes the stable perception information

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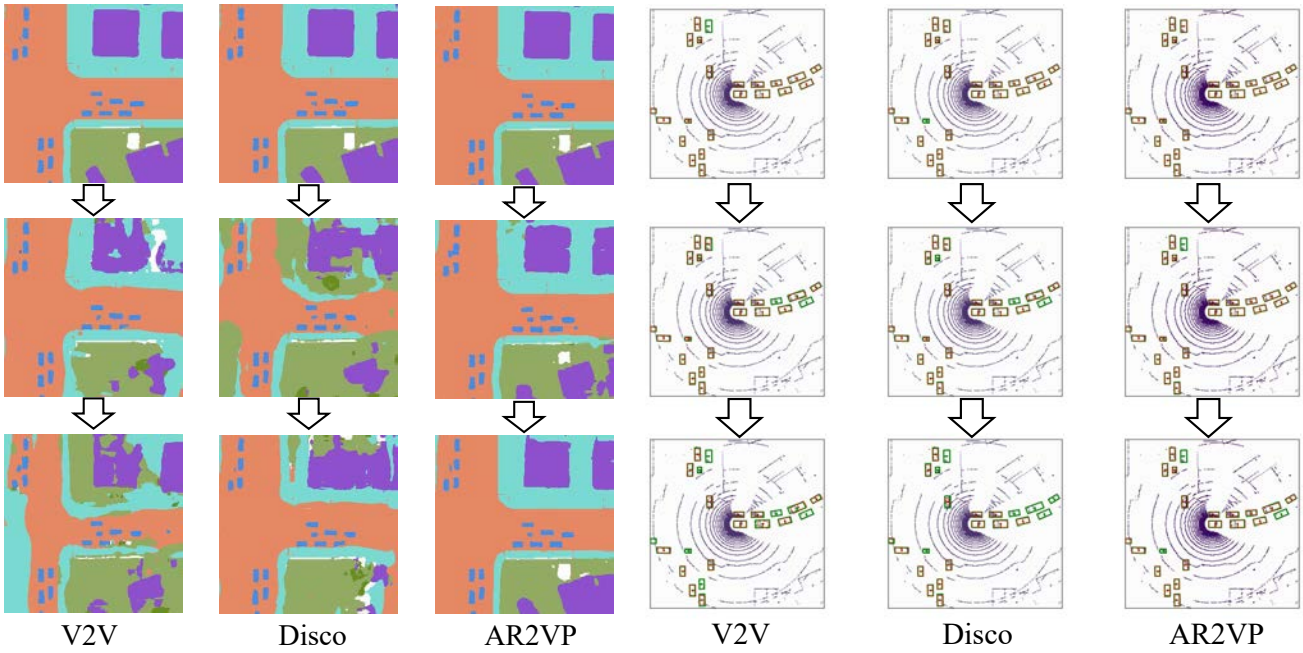

(a) Visualizations of collaborative BEV Seg. on inhibition of forgetting. (b) Visualizations of collaborative BEV Det. on inhibition of forgetting.

Fig. 9: Visualizations of collaborative BEV on inhibition of forgetting.

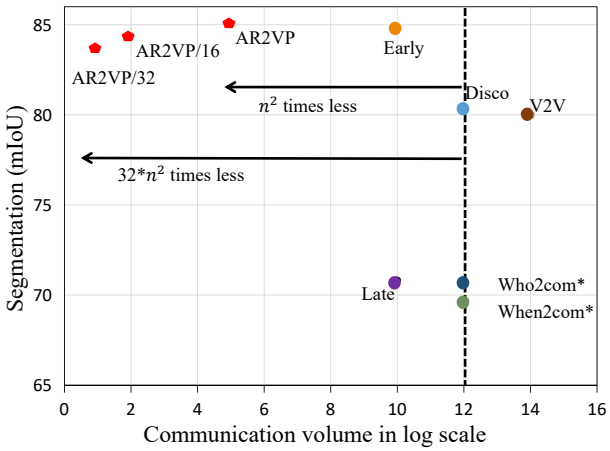

Fig. 10: Dynamic Scene Advantage of RSU. RSU demonstrates stability in perception and possesses geographical advantages in inter-scene changes, as opposed to the continual mobility of vehicles in intra-scene dynamics. With storage and communication capabilities, RSU stores old scenes data and model, enabling adaptation to inter-scene changes.

from RSU to effectively compensate for vehicle perception. Moreover, the compensator benefits from the presence of RSU, showing a positive effect. In the absence of RSU, using vehicles to compensate for other vehicles' perceptions would lead to negative consequences.

Also, we conduct ablation studies on the initial quantity of prompts and the storage capacity of RSU. The results are shown in Table V. Our findings indicate that (1) An inappropriate quantity of prompts makes it challenging to adequately capture the features of the entire scene, leading to the extraction of representative samples that are ineffective in suppressing inter-scene distribution drift issues. (2) Theoretically, the model's robustness is improved with an increase in RSU storage capacity.

V. Conclusion

In this paper, we propose a vehicle-road cooperative perception model, named AR2VP, which is capable of adapting to dynamic environments. It mainly consists of a DPR module, an R2VPC module, and an RPR method. The DPR module efficiently integrates vehicle perceptions to comprehensively capture dynamic factors within the scene, enhancing the perception capabilities of the collaborative perception model. The R2VPC module is geared towards effectively retaining the RSU perception information, especially in the face of intra-scene changes. The RPR method integrates within the RSU's storage capacity and facilitates the retention of a small volume of historical scene data. This approach ensures that the cooperative model maintains a certain level of robustness when confronted with inter-scene changes.

Comprehensive experiments demonstrate that AR2VP achieves adaptability to dynamic environments and an appealing performance-bandwidth trade-off through a more direct design principle.

Limitation and future work: In our experiments, we find that AR2VP is not very robust to small objects, which may be failed recognition or forgotten. We believe the issue is raised by the perception conflicts from different vehicles. In the future, based on our experimental findings, we intend to enhance the AR2VP's capability to recognize small objects. This will involve further refinement of the model to ensure more accurate and effective identification of small entities.

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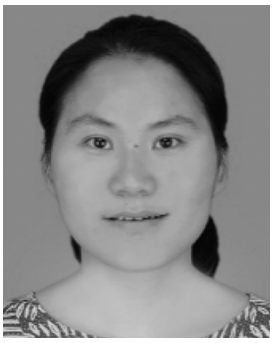

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